NDLI Club

NATIONAL DIGITAL LIBRARY OF INDIA (NDLI)

Central Administration Office • Indian Institute of Technology Kharagpur

OFFICIAL OPERATIONAL HANDBOOK & USER MANUAL

Edition v4.1 Enterprise • October 2026

NDLI Club Management & Regional Employee Activity Tracking System

A Unified, Enterprise-Grade Operational Guidebook for Central Administrative Governance, 7 Regional Nodal Zone Workspaces, 50,000+ Scalable Database Architecture, 200 Real-Time Concurrency, Instant In-Memory Tab Switching, Global UI Error Shields, and Dual-Layer Emergency Restore.

- ✓ Glassmorphic Master Login Gateway
- ✓ 7 Regional Nodal Zones
- ✓ 36 States/UTs Automated Mapping
- ✓ 50/30/20/10 Star Rating Matrix
- ✓ AI Strategic Decision Engine
- ✓ Automated 7-Day Rolling Backups
- ✓ Dynamic Renewal Certificates
- ✓ Tool Box — 12 Utilities
- ✦ 50k+ DB & 200 Concurrency
- ✦ Instant 0ms Tab Caching
- ✦ Global UI Error Boundary
- ✦ Robu AI Assistant
- ✦ Public QR Verification

DOCUMENT RELEASE

Enterprise Handover Edition v4.1 • October 2026

SYSTEM DEPLOYMENT TARGET

Cloud Web Service & Google Drive / GAS Hybrid Architecture

LEAD SYSTEM ARCHITECT & FULL-STACK DEVELOPER

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1. Preface & What's New in Enterprise Edition v4.1

✦ What's New in v4.1 — Rounds 4–8 Consolidated: Hardening, Robu, Reporting & Public QR Verification (2026)

This edition consolidates the platform's 2026 development rounds into one certified operational handbook: (1) **Round-4 Security Hardening:** automated brute-force sign-in lockout, relay-key authenticated server-to-server sync transport, and validated certificate-signature uploads (§5.1); (2) **Round-5 UI Overhaul & Restore Rework:** unified design tokens, motion system and toast notifications across both portals, and an Emergency Restore re-engineered to verify an active administrator credential (§7.9, §7.11); (3) **Round-6 Relay Fix & Robu Assistant:** the Google Drive relay was stabilized and Robu, the floating NDLI Club AI assistant, was introduced — read-only, role-scoped, answering with cited sources (§6.7); (4) **Round-7/8 Reporting & Sample-C Shell:** the one-click Comprehensive Report (PDF & DOC export) with field-difficulty fairness framing (§7.6), and the Sample-C portal shell — a fixed left sidebar with a slim topbar (§7.11); (5) **Public QR Verification:** renewal certificates carry a cryptographically signed QR code that anyone can verify without an account (§7.10.3); (6) **Employee Self-Service Certificates:** regional officers generate renewal certificates and manage the Principal-Investigator signature themselves (§6.8, §7.10.2); (7) **Backup Resilience:** 6-archive rolling retention, signature files bundled into Drive archives, and point-in-time Google Drive copies before destructive actions with drive_sync status reporting (§7.8); (8) **Privacy & Data Handling:** a plain-language charter on what is stored, who may see it, and how long it is kept (§9).

✦ Enterprise Scalability & High-Throughput Upgrades (previous release — retained for reference)

The preceding client-handover edition certified enterprise capacity and fluid operational ergonomics: **(1) 50,000+ Club Database Scaling** with in-memory hash indexing ($O(1)$ sub-millisecond retrieval) and fast-path append; **(2) 200 Real-Time Concurrent Task Capacity** tested under high-throughput HTTP queue stress with 0 errors; **(3) Instant 0ms Tab Switching & In-Memory Caching** in the Employee Portal (Universal Search and Node Activity Logs auto-load without manual refresh); **(4) Resilient Global UI Error Boundary** intercepting client glitches with graceful inline recovery bars; **(5) Complementary Navigation Highlighting** with zero-vertical-scroll employee login and side-by-side portal links; and **(6) Dual-Layer Emergency Codebase & Database Restore Utilities** (CLI, batch, and GUI).

The **National Digital Library of India (NDLI)** Club initiative, coordinated by the **Indian Institute of Technology Kharagpur (IIT Kharagpur)** under the Ministry of Education, Government of India, empowers educational institutions across secondary schools, universities, and research institutes throughout all 36 States and Union Territories.

To support institutional onboarding, regional verification, and continuous field engagement, the **NDLI Club Management and Employee Activity Tracking System** was architected and developed by **Dr. Anirban Mukherjee** as an enterprise-grade operational backbone.

1.1 Core Architectural Principles

- **Enterprise Scalability Engine:** Engineered with sub-millisecond $O(1)$ hash indexing, fast-path disk appends, zero-allocation aggregations, and 50-row pagination, handling 50,000+ records and 200 real-time concurrent requests seamlessly.
- **Zero External Dependency Kernel:** Engineered on Python's standard library (HTTP server, JSON, and CSV engines). Runs natively on any Linux/Windows server or cloud container without dependency friction.
- **Dual-Portal Governance:** Clear logical separation between the Central Administration Master Portal at IIT Kharagpur and the 7 Regional Zonal Workspaces across India.
- **Node-to-Master Synchronization:** Regional officers log field activities and club approvals into dedicated node CSV files, which are immediately reconciled into central master databases with atomic mutex locking.
- **Autonomous Data Resilience:** Automated rolling backups execute every 7 days, maintaining a strict 6-archive rolling retention (maximum 6 archives; older ones pruned) with Google Drive cloud mirroring (§7.8).
- **Modern Minimalist UI Design:** Built with Manrope variable-weight typeface, glass morphism surfaces, amber CTA accents, and 6-level elevation depth.

1.2 Scope & Intended Audience

This operational handbook provides authoritative guidance for Central Administrative Officers at IIT Kharagpur, the 7 Regional Zonal Officers managing daily zonal operations, and DevOps engineers overseeing cloud deployments.

📌 Institutional Attribution & Brand Identity

The application adheres strictly to official IIT Kharagpur branding, utilizing deep forest green tones (`#103125` , `#1A503E` , `#0D7751`) accented by warm amber (`#D97706`) and sky-blue live clock telemetry (`#38BDF8`). Verified developer attribution to Lead Architect **Dr. Anirban Mukherjee** appears in system metadata, headers, certificates, and endpoints.

2. Feature Highlights & Executive Overview

The platform integrates an enterprise-grade suite of administrative governance and high-capacity execution capabilities:

⚡ 50,000+ Club Scalable Engine ENTERPRISE

Sub-millisecond O(1) hash indexing (0.64ms lookups), fast-path row append (13ms), zero-allocation aggregations, and 50-row pagination handling 50k+ records smoothly.

🚦 200 Real-Time Concurrent Requests ENTERPRISE

Threaded HTTP architecture (512 queue slots), atomic RLock thread shields, and bounded upload semaphore stress-tested at 200 concurrent tasks in 179ms (0 errors).

⚡ Instant 0ms Tab Switching & Cache ENTERPRISE

In-memory preloading for Universal Search (Tab 2) and Node Activity Logs (Tab 3). 0ms tab switching, instant filtering (<35ms), and automatic connection retry.

🛡️ Global UI Error Boundary & Restore ENTERPRISE

Global client error interceptors, 10-tier Windows lock backoff, HTTP 500 crash isolation, and one-click codebase restore utilities (restore_to_stable_v2.bat).

🏛️ Central Administration Dashboard

Real-time KPI telemetry, Donut and Trajectory charts, Renewal Attention centers, and multi-dimensional scope filtering by Zone and Academic Year.

🔐 Glassmorphic Master Login Gateway

Deep-frosted glass card with NDLI orbit logo animation, live IST clock, CAPTCHA verification, amber unlock button, and zero-vertical-scroll employee view.

🧰 Integrated Tool Box (12 Utilities)

12 one-click utility shortcuts: osTicket (club-support.ndl.gov.in), 8 Google Workspace services (Gmail, Meet, Drive, Docs, Sheets, Slides, Keep, YouTube), Zoom, Teams, StreamYard.

⭐ Employee Performance Star Matrix

Statutory 50/30/20/10 weighted formula evaluating regional officers with dynamic 1-to-5 star ratings, filtered by Year, Month, and State.

🗺️ National Penetration Roster

Exhaustive roster covering all 36 States & UTs with real-time club density, color-coded health indicators, and no-horizontal-scroll layout.

🧠 AI Strategic Decision Engine

Heuristic intelligence formulating policy recommendations, identifying penetration deficits, and generating 30/90/365-day milestone roadmaps.

📜 Dynamic Verification Certificates

Generates official IIT Kharagpur Renewal Certificates embedding seals, dynamic signatories, Sutradhar typography, instant vector PDF export, and a publicly verifiable QR code (§7.10.3).

☁️ Google Drive Cloud Mirroring

Automated cloud mirroring of 7-day and manual ZIP backups directly into Google Drive with strict 6-archive rolling retention and point-in-time copies before destructive actions (§7.8).

🤖 Robu — AI Club Assistant V4.1

A floating assistant in both portals answering live-data and documentation questions — read-only, role-scoped, with numbered citations, deep links, and proactive nudges (§6.7).

✅ Public QR Verification V4.1

Renewal certificates embed a signed QR token: anyone holding the printed certificate can verify it online — no account needed, no search facility, no personal data exposed (§7.10.3).

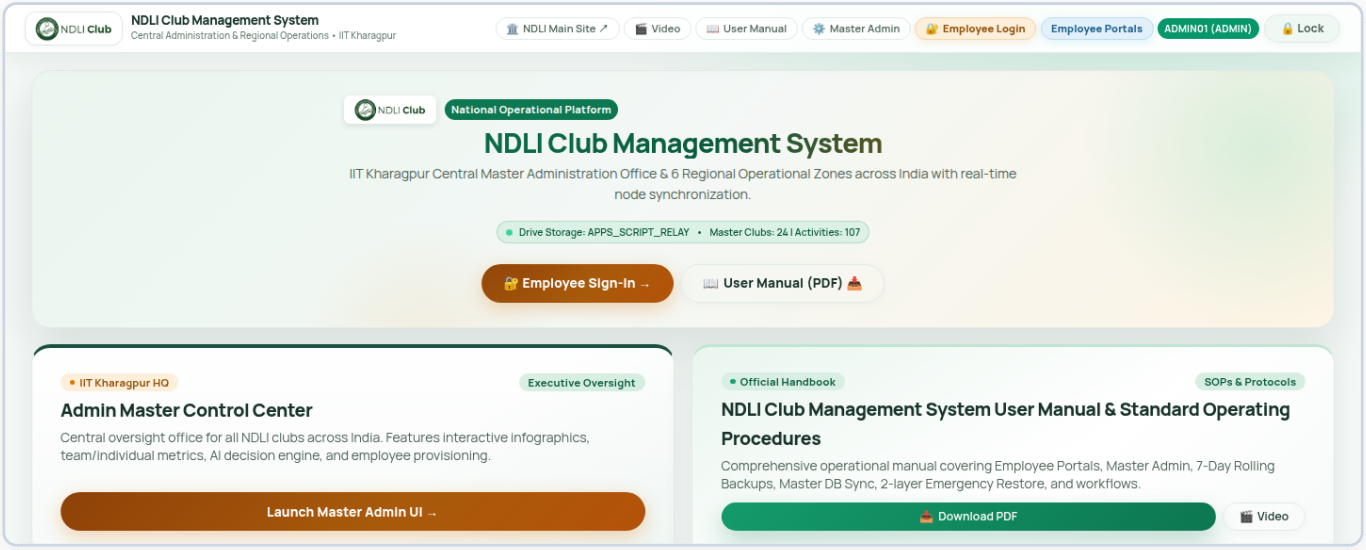


Figure 2.1: Unified Landing Portal — Modernized Card Grid with Admin Office and 7 Regional Zonal Workspaces

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4. Detailed Specifications & Database Schemas

The platform is powered by a high-performance, atomic CSV database engine. Every operation is schema-validated to guarantee consistency across master and employee node tiers.

4.1 Master Database Schema Specifications

Database File	Schema Columns	Operational Function
`master_clubs.csv`	`id, reg_no, name, state, zone, patron_email, president_email, secretary_email, approved_by, approval_date, last_renewal_date, renewal_date`	Stores all active NDLI Clubs nationwide. Enforces strict +1 year renewal invariants from the previous due date (rolled forward when in the past).
`master_activities.csv`	`id, emp_id, timestamp, support_type, notes, state, zone`	Aggregated log of all field interactions (calls, ticket closures, webinars, offline campus visits) across India.
`master_quotas.csv`	`emp_id, emp_name, zone, clubs_approved, total_activities, last_activity, updated_at`	Scoreboard tracking individual employee performance quotas and daily logging volume.
`master_users.csv`	`id, email, full_name, role, zone, assigned_states, is_active, created_at, password_hash, salt`	Credentials database. Stored only as one-way PBKDF2-HMAC-SHA256 digests (100,000 rounds, per-user 16-byte random salts).
`master_issues.csv`	`id, club_id, club_name, emp_id, reported_at, issue_notes, status, next_reminder_date, resolved_at, resolution_notes`	Engine for 72-hour employee reminders, 7-day recurrent cycles, and 30-day Admin escalations.

4.2 Regional Zonal Jurisdiction Roster

User ID	Regional Officer	Zone	Territorial Jurisdiction (36 States & UTs)
`EMP01`	**Rohan Sharma**	NORTH	Jammu & Kashmir, Ladakh, Uttarakhand, Himachal Pradesh, Chandigarh, Punjab, Haryana, Delhi, Uttar Pradesh
`EMP02`	**Pooja Verma**	CENTRAL	Madhya Pradesh, Chhattisgarh
`EMP03`	**Amit Patel**	WEST	Rajasthan, Gujarat, Maharashtra, Goa, Daman and Diu, Dadar & Nagar Haveli
`EMP04`	**Debabrata Ghosh**	EAST	Bihar, Jharkhand, West Bengal, Odisha
`EMP05`	**Mayanglambam Singh**	NORTH EAST	Sikkim, Assam, Arunachal Pradesh, Meghalaya, Manipur, Tripura, Nagaland, Mizoram
`EMP06`	**K. Venkatesh**	SOUTH	Andhra Pradesh, Telangana, Karnataka (South Sector 1)
`EMP07`	**Ananya Nair**	SOUTH	Tamil Nadu, Puducherry, Kerala, Andaman & Nicobar Island, Lakshadweep (South Sector 2)

> ℹ️ **Node Database Structure**
> Each officer maintains a dedicated local node database directory at `data/employees/empXX/` containing `clubs.csv` (their authorized clubs) and `activity_log.csv` (their daily interaction log). All entries are automatically synchronized to the Central Master databases via the Node-to-Master Sync Engine.

5. Master Login Gateway — Enterprise Secure Access

> **🔐 Master Login Gateway Overview**
> The Master Login Gateway is the primary security barrier for the entire NDLI Club Management System. It presents a full-viewport **glassmorphic login screen** when accessing the root URL (`/`). The gateway requires verification of administrator identity before revealing the central portal selector. Regional Employees bypass this screen by navigating directly to `/employee` .

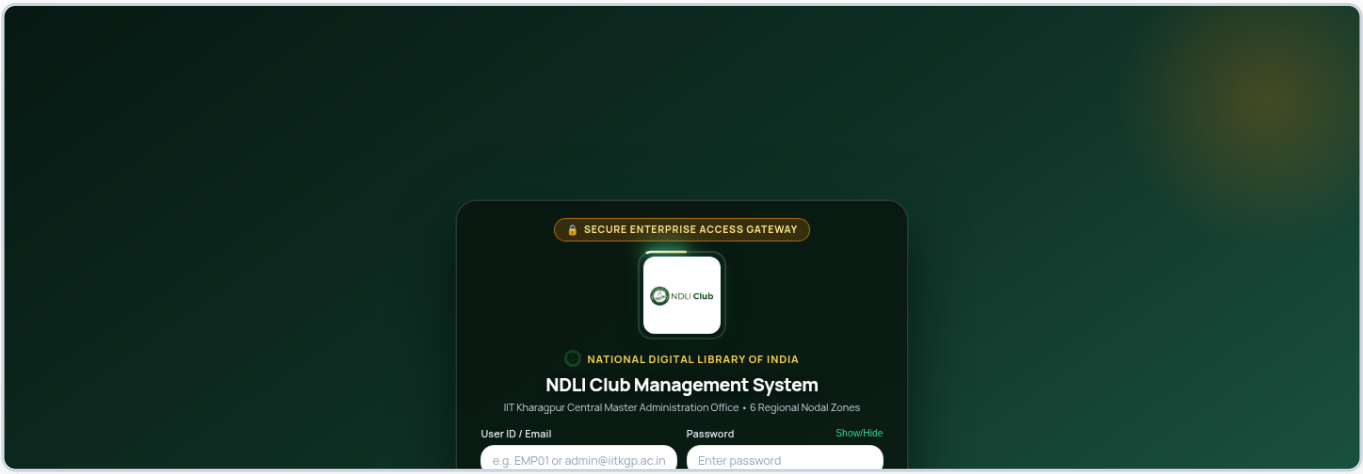


Figure 5.1: Master Login Gateway — Full-Viewport Glassmorphic Security Screen with NDLI Orbit Rings, Live IST Clock & CAPTCHA

5.1 Security Architecture & CAPTCHA Protection

The Gateway employs a layered visual and cryptographic security architecture:

- **PBKDF2-HMAC-SHA256 Password Hashing** — account credentials are stored only as one-way digests (100,000 rounds with per-user 16-byte random salts); usable secrets are never kept.
- **Dynamic Canvas Anti-Bot CAPTCHA:** A 5-character randomized alphanumeric challenge is rendered on an HTML5 canvas with random distortion angles, lines, and background noise. It regenerates on each reload or refresh click.
- **Cryptographic Session Tokens:** Upon authentication, a secure 24-byte random session token is stored in `sessionStorage`, automatically expiring when the browser tab is closed.
- **Synchronous Auth Pre-Unlock:** Legitimate ongoing sessions bypass the lock screen instantly via a synchronous script guard, preventing any jerky screen flash during portal navigation.
- **Automated Brute-Force Lockout:** 8 failed sign-in attempts for the same identifier inside a 10-minute window trigger an automatic 5-minute lockout (the server answers HTTP 429, "Too many failed sign-in attempts"), throttling credential-guessing attacks. While the lockout lasts, sign-in is simply refused — after the 5-minute cool-off the same credential signs in normally, with no administrator action required.
- **Transport & Upload Hardening:** the server-to-server sync/backup transport is relay-key authenticated, certificate signature uploads are validated (image magic bytes, 2 MB cap), and file downloads carry short-lived bearer tokens with automatic stale-token cleanup.

5.2 Step-by-Step: Unlocking the Master System Access

1. **Access Root URL:** Open your browser and navigate to `https://ndli-club.onrender.com/` . The glassmorphic lock card appears centered over the forest-green backdrop.
2. **Enter Master Identifier**: input the registered administrator email address issued to you by your administrator (this handbook deliberately publishes no example identifiers).
3. **Enter Master Password**: type the credential issued to you by your administrator into the password field. Use the **Show/Hide** toggle to confirm what you typed.
4. **Solve Anti-Bot CAPTCHA:** Read the 5-character code in the security canvas. Type the code (case-insensitive) in the input box. Click ↻ **Refresh** if characters are difficult to read.
5. **Unlock Access:** Click the amber gradient **"🔓 Unlock System Access →"** button. The lock screen smoothly fades away in 0.18s, revealing the full Landing Portal.

6. Regional Employee Portal Operations

The **NDLI Club Employee Portal** is the operational workspace for the 7 Regional Zonal Officers. It features real-time clock synchronization, pending reminder feeds, and structured data entry workflows.

6.1 Employee Login, Quick-Select Chips & Zero-Scroll Ergonomics

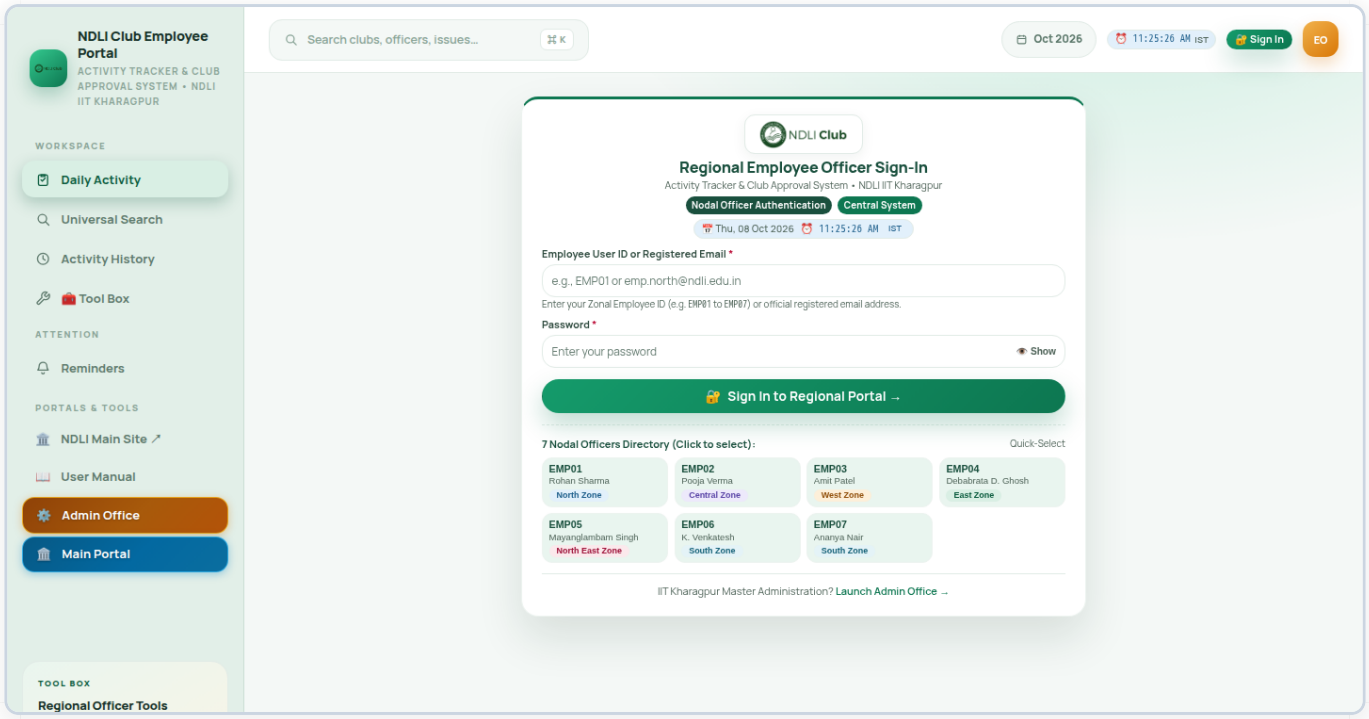


Figure 6.1: Regional Employee Sign-In — Interactive Quick-Select Officer Chips Directory with Live IST Clock

1. **Navigate to Portal:** Open `/employee` or click **"Sign In to Portal"** on any EMP card in the main landing page.
2. **Quick-Select Regional Officer Chip:** Click any of the 7 officer chips (e.g., *EMP01 — Rohan Sharma — North Zone*). The User ID is automatically filled into the login form.
3. **Credential Authentication:** Enter the credential issued to you by your administrator into the password box and click **"Sign In to Regional Portal →"**.
4. **Workspace Activation:** An encrypted 24-byte session token is generated, loading the officer's dedicated workspace with their node database pre-cached.

> ℹ️ **Zero-Scroll Ergonomics & Direct Node Shortcuts**
> The login interface features equalized officer chip grids, compact field spacing, and zero vertical scrolling on standard displays. Officers can also navigate directly via node shortcuts: `/employee/EMP01` through `/employee/EMP07`. The Master Gateway's brute-force lockout applies here too: 8 failed attempts inside 10 minutes pause sign-in for 5 minutes (HTTP 429) — wait for the cool-off and sign in again with the credential issued to you by your administrator.

6.2 Step-by-Step: Logging Daily Support Activities (SEC A)

Zonal officers engage with educational institutions daily to resolve obstacles and conduct orientation sessions. Every interaction must be logged immediately in **SEC A — Support Log**:

1. **Open Daily Activity Tab:** Confirm you are on the **"Daily Activity"** tab of the Employee Portal.

2. **Select Nature of Support:** Choose the appropriate activity tier:
   - **Phone call and remote assistance:** Voice calls, WhatsApp consultations, and remote desktop setup.
   - **Closing of OS Ticket:** Formal resolution of institutional support tickets logged via NDLI helpdesk.
   - **Online training:** Conducting virtual orientation webinars, faculty workshops, and student onboarding.
   - **Offline training:** On-premise physical campus visits, seminars, and library workshops.

3. **Activity Notes:** Enter brief notes identifying the institution, key stakeholders, and topics addressed.

4. **Submit Entry:** Click **"Submit Activity Log"**. The entry is recorded in `activity_log.csv` with automatic timestamping and synced incrementally to the Central Master Database.

6.3 Step-by-Step: Approving & Onboarding New NDLI Clubs (SEC B & C)

When an institution completes its registration requirements, authorize the club through **SEC B & C**:

1. **Trigger Approval Form:** Click the button **"⚡ SEC B: Approve New NDLI Club"** to reveal SEC C with a smooth spring animation.

2. **Input Institutional Codes:** Enter unique Club ID (format: `CLUB-[STATE]-[YEAR]-[NUM]`), official Registration Number, and Institution Name.

3. **State Selection (Auto-Zone Engine):** Select the institution's state from the dropdown. The system automatically determines and locks the statutory Zone (North, Central, West, East, North East, South), preventing territorial misclassification.

4. **Stakeholder Contact Matrix:** Provide verified email addresses for Patron (Head of Institution), President (Faculty Lead), and Secretary (Student Representative).

5. **Authorize Club:** Click **"✓ Confirm & Authorize New NDLI Club"**. The club is committed to the node database, increments the officer's quota, and syncs to `master_clubs.csv`.

> ⚠️ **Mandatory Next Renewal Date Invariant (Rule 10)**
>
> Upon club approval, the system enforces **Rule 10**: the upcoming renewal date is automatically set to *exactly 1 calendar year (+1 year)* from the **previous renewal due date** (keeping the club's anniversary from the Date of Approval). If a club is renewed late/overdue, the due date is rolled forward in 1-year steps until it lies in the future. Under no circumstances can a renewal date precede DOA.

6.4 Universal Search, 0ms In-Memory Cache & 50-Row Pagination

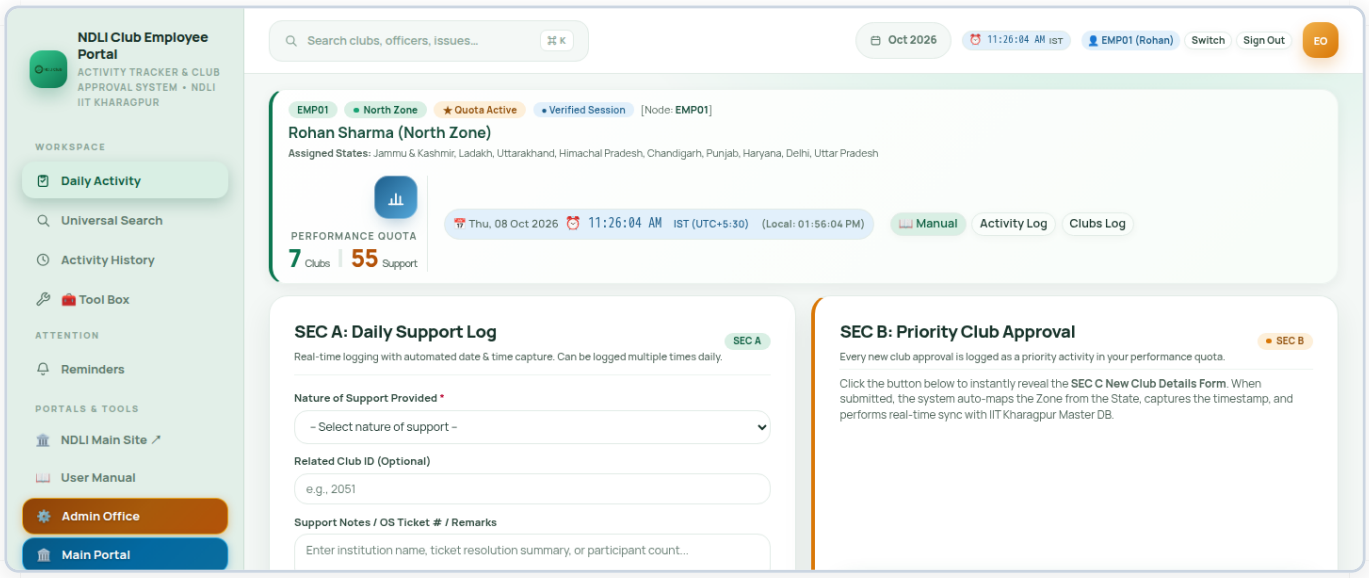


Figure 6.2: Employee Workspace — Universal Search & Edit Engine with In-Memory Caching and 50-Row Pagination

1. **Instant In-Memory Cache:** The entire club registry is pre-cached client-side (`ALL_CLUBS_CACHE`) upon login. Switching to Tab 2 renders in **0ms** without network latency.
2. **Sub-35ms Search & 50-Row Pagination:** Typing in the query box filters records instantly in under 35ms. Smooth pagination controls (First, Prev, Next, Last) preserve fluid UI frame rates even with thousands of clubs.
3. **Inspect, Edit & Annual Renewal:** Click **"Inspect / Edit"** to modify stakeholder emails, or click **"🔄 Process Renewal"** to advance renewal by +1 year (from the previous due date) and export the print-ready renewal certificate — which embeds the public verification QR code (§7.10.3). Certificate generation and Principal-Investigator signature management are detailed in §6.8.

6.5 Node Activity Logs (Tab 3) Auto-Loading & 72-Hour Reminders

Instant Activity Rendering: Navigating to *My Node Activity History* automatically pre-renders from `ALL_ACTIVITIES_CACHE` with 0ms delay (no manual refresh needed). An animated loader appears during live updates, with an inline `↻ Retry` button for connection resilience.

- **72-Hour Grace & 7-Day Reminders:** Reported club issues display active alert banners 72 hours post-logging, recurring weekly until marked **"Resolved"**.
- **Node CSV Exports:** Officers can click **"Download Activity Log"** and **"Download Clubs Log"** to export instantaneous local CSV archives.

6.6 Integrated Tool Box — 12 Operational Utilities V4.1 FEATURE

The **Tool Box** tab in the Employee Portal centralizes all daily external services used by Regional Zonal Officers into a single-click launcher. Tools are strategically ordered: operational helpdesk first, followed by Google Workspace services, and concluding with virtual communication platforms.

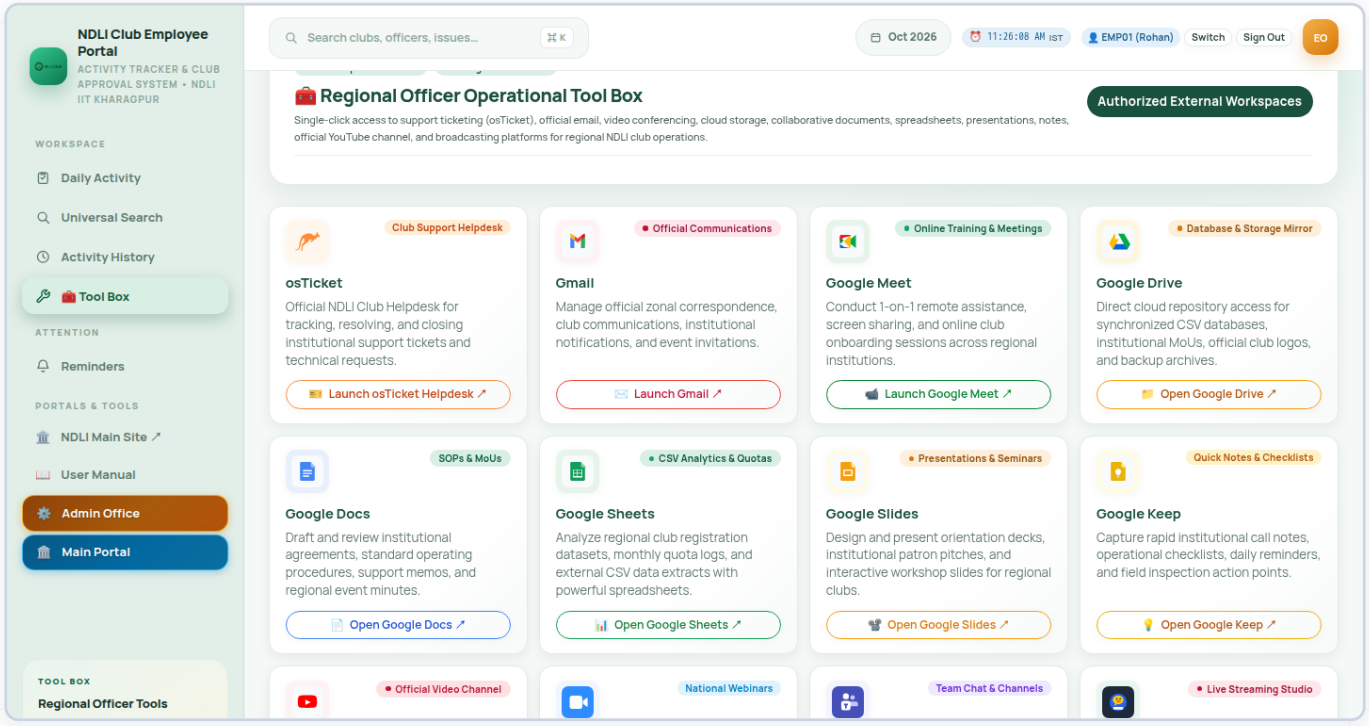


Figure 6.3: Integrated Tool Box Tab — 12 Production Utilities Arranged by Operational Priority (osTicket First, Google Suite, Broadcast)

Complete Tool Directory & Functional Purpose

🎟 PRIORITY HELPDESK
osTicket
Official NDLI Club Helpdesk (`club-support.ndl.gov.in`). Positioned first for swift issue ticket resolution.

📧 GOOGLE WORKSPACE
Gmail
Official institutional correspondence, stakeholder notifications, and certificate dispatches.

📹 GOOGLE WORKSPACE
Google Meet
High-definition virtual meetings for online institutional orientation webinars and training.

☁️ GOOGLE WORKSPACE
Google Drive
Cloud repository for regional club files, presentation decks, and collateral templates.

📄 GOOGLE WORKSPACE
Google Docs
Collaborative drafting of institutional MoUs, official letters, and inspection notes.

📊 GOOGLE WORKSPACE
Google Sheets
Tabular management of state-wise target lists, regional rosters, and institutional contacts.

📑 GOOGLE WORKSPACE
Google Slides
Presentation decks for club orientation seminars and state-level education workshops.

📝 GOOGLE WORKSPACE
Google Keep
Rapid task checklists, field visit notes, and follow-up memos during campus travel.

▶️ GOOGLE WORKSPACE
YouTube
NDLI video tutorials, user walkthroughs, and national event recordings repository.

💬 VIRTUAL CONFERENCING
Zoom
Large-scale webinars, multi-institution digital conclaves, and interactive Q&A sessions.

💼 ENTERPRISE COLLABORATION
Microsoft Teams
Government stakeholder meetings, inter-ministry committees, and regional coordination.

🎙 LIVE BROADCAST
StreamYard
Multi-stream live broadcasting across social channels for nationwide NDLI Club launches.

6.7 Robu — Your NDLI Club AI Assistant V4.1 FEATURE

Both portals include **Robu**, a floating AI assistant (bottom-right button) that opens a chat panel. Robu is named after the resourceful robot from Satyajit Ray's *Professor Shonku* stories — built modestly, worth considerably more. She answers in plain language, shows her sources, and never touches your data.

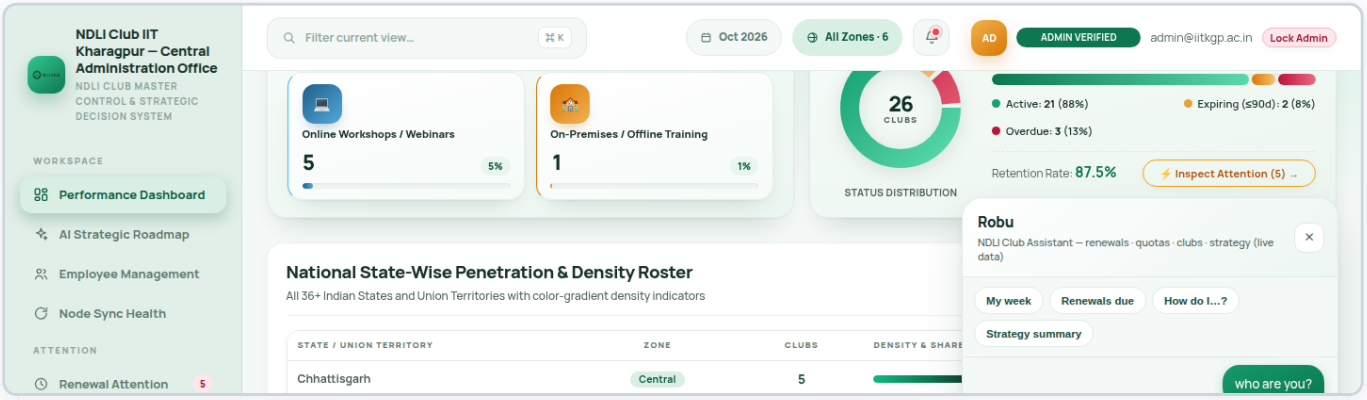


Figure 6.4: Robu in Action — the Floating Launcher Opens the Chat Panel (shown answering "who are you?" with her Professor Shonku origin story)

What Robu Answers & Does

- **Live data answers (role-scoped, read-only):** "How many clubs did I onboard this month?", "Which renewals need attention?", open issues and reminders, quota standings, activity history — answered from your own portal data. Regional officers see ONLY their own node's records; administrators see the national picture. Robu is strictly read-only: she never creates, edits, or deletes anything.
- **Documentation answers with citations:** "How do I approve a renewal?" — answered from this manual and the Technical Reference Manual, with numbered source citations so you can verify the section yourself.
- **Deep links (one-tap navigation):** when your question maps to something you can act on, Robu offers a suggestion card that opens the right screen for you — the renewal-attention list, the issue-reminders queue, a club's record — on the portal you are already using.
- **Strategic report explainers:** ask "explain the report" (or "explain recommendation 2") and Robu walks through the AI Strategic Decision Module's recommendations in plain language — what each one means and what to do about it — grounded in live numbers.
- **Proactive nudges:** a small badge appears when something needs attention — renewals overdue or due within 14 days first, then 30-day escalations, then an officer quota at risk. One nudge at a time, quiet hours 21:00–08:00, dismissed once seen (it returns only if the flagged set changes). Read-only: Robu informs, never acts.

Honest Limits, Learning & Safety

- **Honest limits + template mode:** out-of-scope questions get a plain "I can't see that" instead of a guess. If no AI language-model key is configured on the server, Robu runs in clearly-labelled template mode — deterministic answers built from live data. Still useful, never fake.
- **Feedback & learning:** every answer carries 👍/👎 buttons. Feedback is recorded, and a feedback-learning loop evaluates and promotes better answers for future questions.
- **Safety rails:** guardrails screen every answer — no invented data, no credential leaks, no actions beyond reading — and an audit trail of assistant interactions is kept for quality and safety review.

> **📌 Privacy — Robu Sees Only What YOUR Login May See**
>
> Questions and 👍/👎 feedback are logged for quality with the platform's standard retention. The full picture of what the platform stores — and who may see it — is set out in §9 Privacy & Data Handling.

6.8 Renewal Certificates & PI Signature (Employee Self-Service) V4.1 FEATURE

Per the client requirement of 2026-09-26, renewal certificates are a **self-service task for regional officers**: certificates are generated from the employee's own workspace, and the Principal-Investigator (PI) signature printed on them is managed by the officer directly. The Central Administration Office retains the standardization controls described in §7.10.2.

1. **Generate the certificate:** open the club's record (Universal Search → **"Inspect / Edit"**) and process the renewal. The print-ready certificate is composed instantly — institutional seals, dynamic signatories, and Sutradhar typography — and exports as a vector PDF. Every certificate embeds the publicly verifiable QR code (§7.10.3).
2. **Manage the PI signature:** the signature image printed on certificates is uploaded or replaced by the officer from the same workspace. Every upload is validated server-side — only genuine PNG or JPEG image data is accepted (magic-bytes check) and the size cap is 2 MB; oversized or non-image files are rejected with a clear message. Each change is attributed to the signed-in officer.
3. **Reprint at any time:** certificates can be regenerated from the club record whenever a reissue is needed. The verification QR token is minted server-side at generation time (§7.10.3), so every reprint carries a genuine, verifiable code.

> ⚠️ **Signature Care**
> The PI signature is an official attestation asset. Only upload the authorized signatory's genuine signature image, keep it current when the signatory changes, and remember that signature files are bundled into backup archives (§7.8) so a restored system remains certificate-ready. How signature assets are handled as data is covered in §9.6.

7. Central Administration Office Portal

The **Central Administration Office Portal** at IIT Kharagpur is the executive master control console. It synthesizes field reports from all 7 regional zones into actionable statistical intelligence.

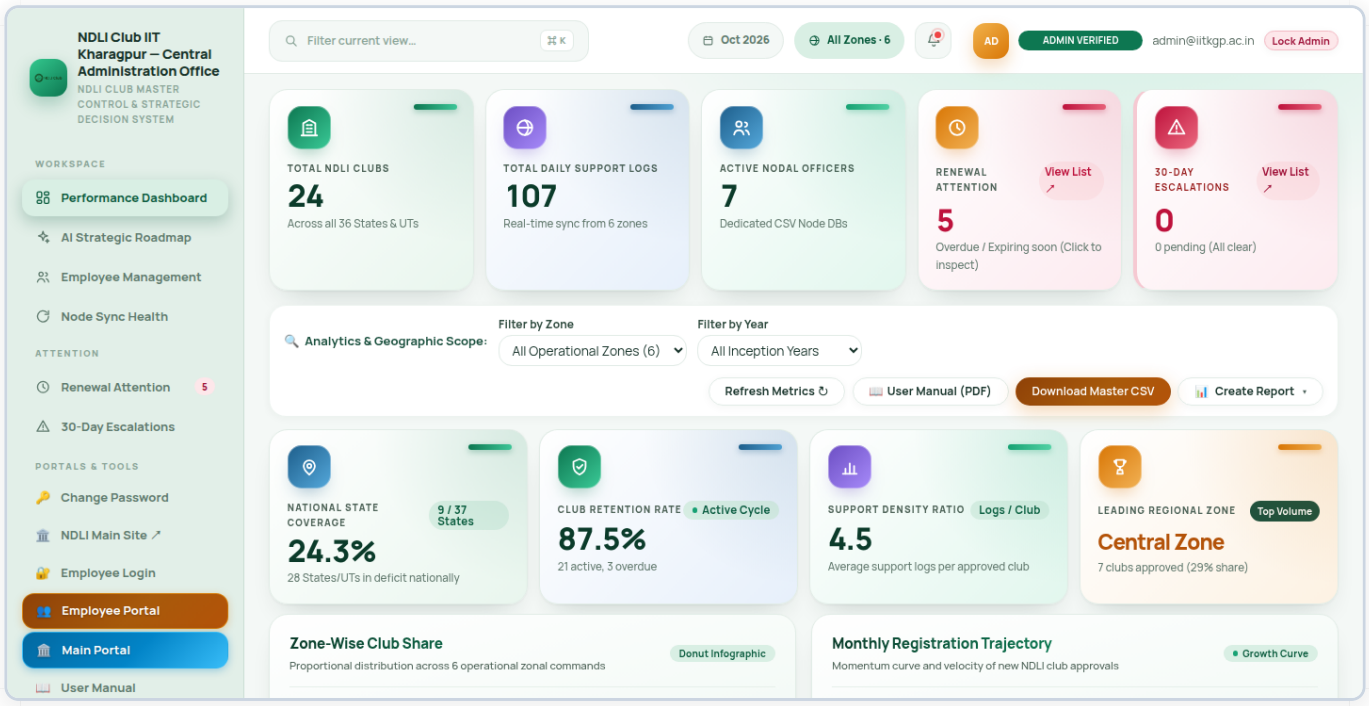


Figure 7.1: Central Administration Executive Dashboard — Real-Time KPIs, Donut Charts & Scope Filters

7.1 Admin Security & Executive KPI Monitors

Administrative access is protected by role-based authentication of the administrator account issued by the Central Administration Office. The dashboard displays five top-level operational meters:

- **Total NDLI Clubs:** Cumulative authorized clubs across all 36 States & UTs.
- **Total Daily Support Logs:** Nationwide count of field interactions submitted across all regional nodes.
- **Active Regional Nodes:** Number of operational regional databases communicating in real-time.
- **Renewal Attention Center:** Highlights clubs with expired or imminent registration deadlines (within 30 days). Clicking the card opens an interactive drill-down inspection list.
- **30-Day Escalations Monitor:** Displays persistent club issues unresolved for 30 or more days across any zone, requiring central administrative intervention.

7.2 Statistical Infographics & Geographic Scope Filtering

The dashboard renders dynamic visual charts representing national adoption patterns:

- **Zone-Wise Distribution Donut Chart:** Displays institutional concentration across the 6 geopolitical zones (North, Central, West, East, North East, South).
- **Monthly Trajectory Area Chart:** Illustrates onboarding velocity month-over-month, showing seasonal registration trends.
- **Support Classification Breakdown:** Tracks the nationwide proportion of remote phone support, OS ticket resolutions, online webinars, and physical campus trainings.
- **Scope Filter Bar:** Administrators can filter all dashboard metrics simultaneously by **Zone** (e.g., "North Zone Only") and **Academic Year** (e.g., 2026, 2025).

7.3 Handling 30-Day Administrative Escalations

Issues unresolved at the regional level for more than 30 days automatically trigger central escalation:

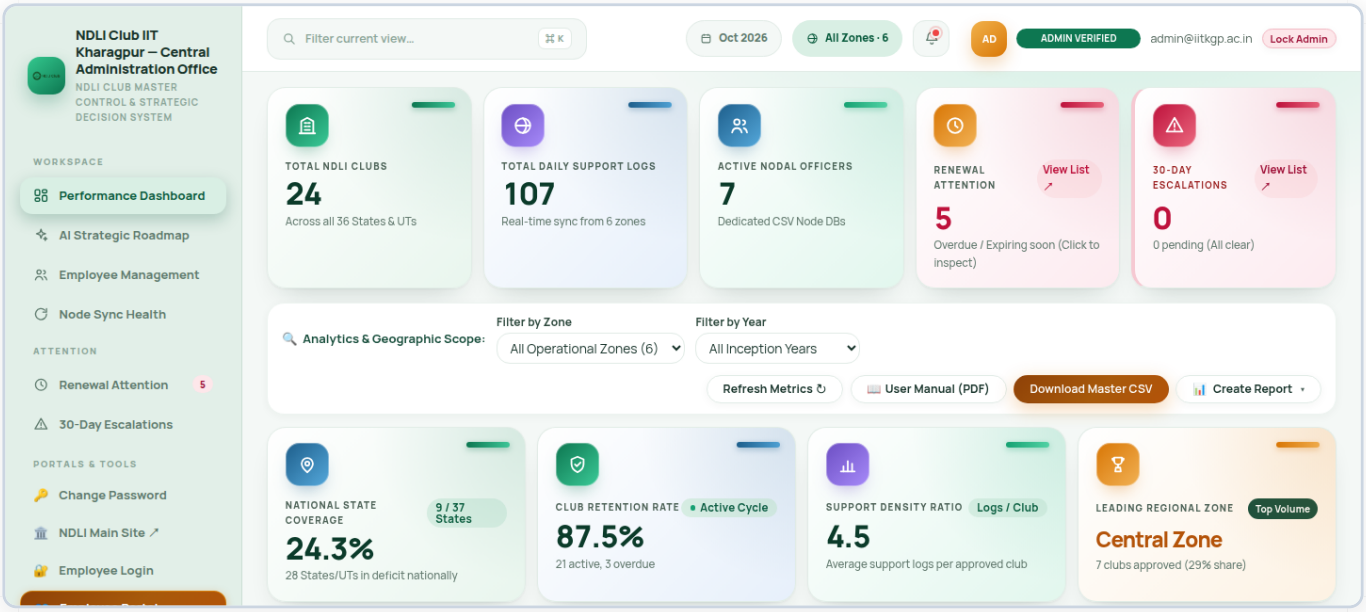


Figure 7.2: Admin Portal Lower Section — Support Activity Breakdown, Renewal & Retention Health, Monthly Registration Trajectory, Zone-Wise Club Share & the Activity Distribution Matrix

1. Click the red **"30-Day Escalations"** card in the KPI header.
2. Review the institution's grievance history, originating zonal officer, and days elapsed.
3. Click **"Review & Resolve"** to record Central Office directives and mark the ticket **"Resolved"** or **"Not Resolved"** (re-queuing for 7-day central tracking).

7.4 Employee Performance Evaluation & Star Rating Matrix

The **Employee Performance Evaluation & Star Rating Matrix** provides an objective merit evaluation of all 7 regional zonal officers based on an approved multi-criteria weighting formula:

```
Composite Score = (Clubs Approved × 0.50) + ((OS Tickets Closed + Phone Calls) × 0.30) + (Offline Training × 0.20) + (Online Training × 0.10)
```

Evaluation Metric	Weightage	Operational Significance
New Club Approvals	50% WEIGHT	Primary organizational KPI reflecting institutional expansion and onboarding capacity.
OS Tickets Closed & Remote Calls	30% WEIGHT	Measures responsiveness to club operational queries and problem-solving velocity.
Offline Field Training	20% WEIGHT	Physical campus workshops, library orientations, and direct faculty development sessions.
Online Training	10% WEIGHT	Virtual webinars, digital orientation sessions, and remote demonstrations.

Star Rating Thresholds & Categorization

Star Rating	Normalized Benchmark	Performance Appraisal Designation
⭐⭐⭐⭐⭐	80% – 100% of Benchmark	**Outstanding:** Consistently exceeds institutional quotas, fast ticket resolution, proactive field engagement.
⭐⭐⭐⭐	60% – 79% of Benchmark	**Commendable:** Strong performance across onboarding, ticket closing, and training.
⭐⭐⭐	40% – 59% of Benchmark	**Satisfactory:** Fulfills standard institutional quotas and baseline activity targets.
⭐⭐	20% – 39% of Benchmark	**Needs Improvement:** Below expected club onboarding velocity or lagging ticket resolution.
⭐	0% – 19% of Benchmark	**Critical Deficit:** Immediate administrative consultation and field support required.

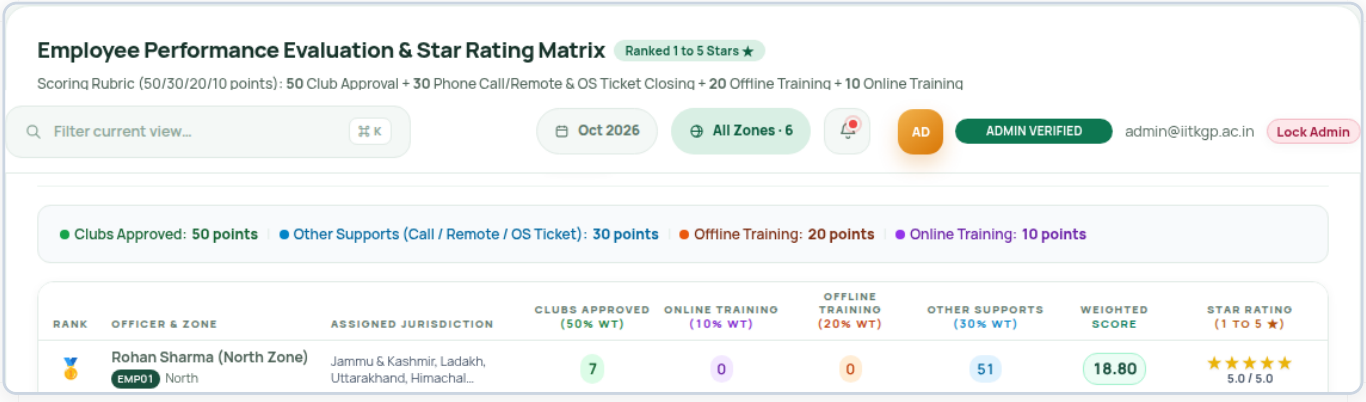


Figure 7.3: Real-Time Employee Star Rating Matrix with Month, Year & State Filters

7.5 National State-Wise Penetration & Density Roster

The **National State-Wise Penetration & Density Roster** renders all 36 States and Union Territories with zero horizontal scrolling, enabling instant nationwide assessment:

- **Territory & Jurisdiction:** Lists each State/UT, assigned Regional Zonal Officer, and geopolitical Zone.
- **Club Density:** Real-time count of active NDLI Clubs operating within that state.
- **Activity Timestamp:** Displays the exact date of the latest onboarding or support engagement.
- **Health Status Badges:** Color-coded indicators denoting High Density (Green), Medium Density (Gold), or Priority Expansion (Red).

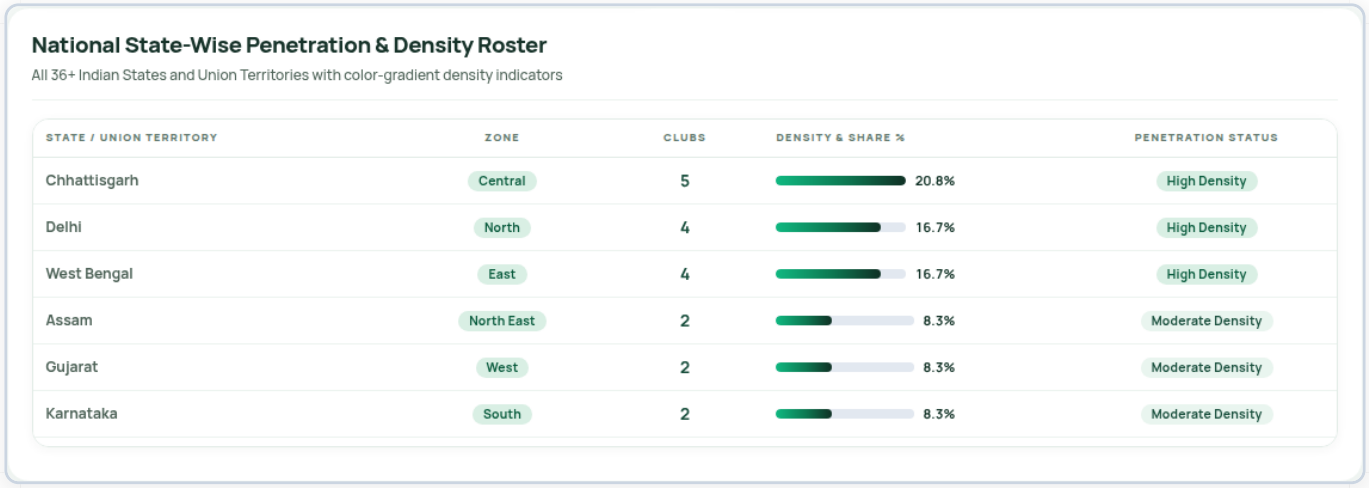


Figure 7.4: National State Penetration & Institutional Density Matrix Covering All 36 States & Union Territories

🗺️ **Geographic Synchronization**

Updates submitted by regional officers in SEC B are immediately mapped to their respective State in this roster. The administrative leadership can click any column header to sort by Density, State name, or Last Interaction Date.

7.6 AI Strategic Decision Module & Policy Roadmaps

The **AI Strategic Decision Engine** executes continuous heuristic evaluation across the aggregated master databases to generate strategic recommendations for central leadership:

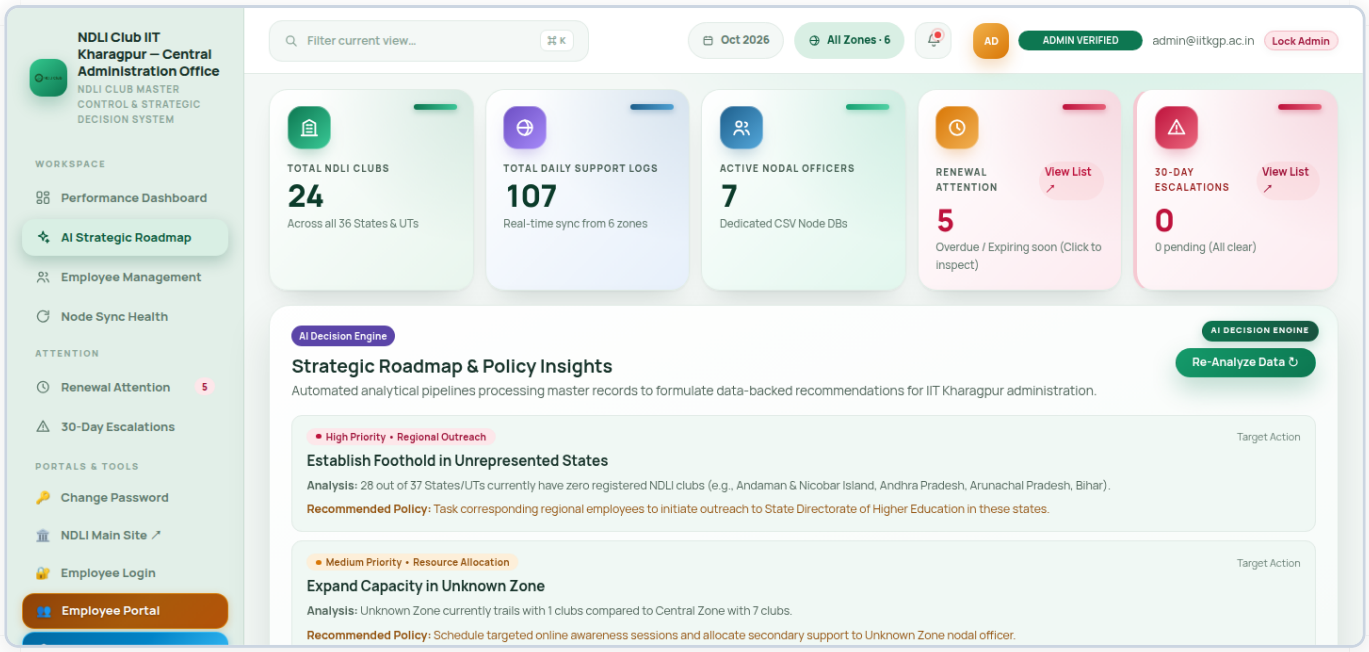


Figure 7.5: AI Strategic Decision Engine — Growth Corridors, Territorial Deficit Formulations & Milestone Roadmaps

- **State Penetration Deficit Formulation:** Automatically identifies territories with club densities below national population-adjusted benchmarks, recommending targeted state-level campaigns.
- **High-Growth Corridor Detection:** Identifies zones exhibiting accelerating adoption momentum to allocate supplemental training materials and regional resources.
- **Milestone Policy Roadmaps:** Synthesizes operational directives into **30-Day Immediate**, **90-Day Mid-Term**, and **1-Year Strategic Milestones**.
- **Automated Resource Allocation Recommendations:** Recommends reallocating training webinars and physical orientation camps to high-potential lagging states based on seasonal academic calendars.

One-Click Comprehensive Report (PDF & DOC Export)

The Admin dashboard also ships a **Comprehensive Report Generator**: the compact **Create Report** button (dashboard toolbar, right-hand side) opens a short **PDF / DOC** menu — pick the format and the full national picture is assembled into a print-ready document covering the executive summary, national snapshot, detailed statistical analytics, the field-difficulty tier split, the AI strategic roadmap (30/90/365-day), employee performance analysis, and generation metadata. The report is the only printable surface while it is open, so exports stay clean and presentation-ready. For a plain-language walkthrough of any recommendation, ask Robu to "explain the report" (§6.7).

> ⚖️ **Fairness Frame — Advisory Layer Only**
>
> The report explicitly accounts for materially harder field conditions: **High-Difficulty Field** tiers cover Jammu & Kashmir, Ladakh, and the eight North-East states (Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura), with clearly-labelled additive sections, an advisory narrative, and dual pacing targets (national pace vs high-difficulty-field pace). The official composite scores, star ratings, and the statutory 50/30/20/10 rubric are rendered verbatim from the platform engines and are never re-weighted — the fairness framing is context, not adjustment.

7.7 Node-to-Master Database Reconciliation Engine

The **Master DB Reconciliation** module (operated via the **"Run Full Node-to-Master Reconciliation"** button in the Admin Office Portal) serves as the primary system-wide data integrity and state recovery mechanism for IIT Kharagpur:

- **Distributed Node Audit:** Recursively scans all regional officer node databases across India (`data/employees/emp01/` through `emp07/`, plus any newly provisioned officers), inspecting their local `clubs.csv` and `activity_log.csv` files.
- **Missing Record Discovery & Merge:** Discovers any club approvals or support activities logged during intermittent network connectivity or restored from historical backups, and seamlessly merges them into the Central Master Database without overwriting newer records or losing data.
- **Schema Normalization & Renewal Recalculation:** Validates all date stamps against strict ISO-8601 formatting, verifies stakeholder contact emails, and recalculates dynamic annual club renewal dates.
- **Quota & Star Rating Synchronization:** Recomputes each officer's approved club counts and support activity tallies directly from raw source records, guaranteeing that `master_quotas.csv` and the **Employee Performance Evaluation & Star Rating Matrix** display 100% verified, tamper-proof ratings.

7.8 Automated 7-Day Backups & Google Drive Cloud Mirroring

The platform incorporates an autonomous background backup engine and disaster recovery console:

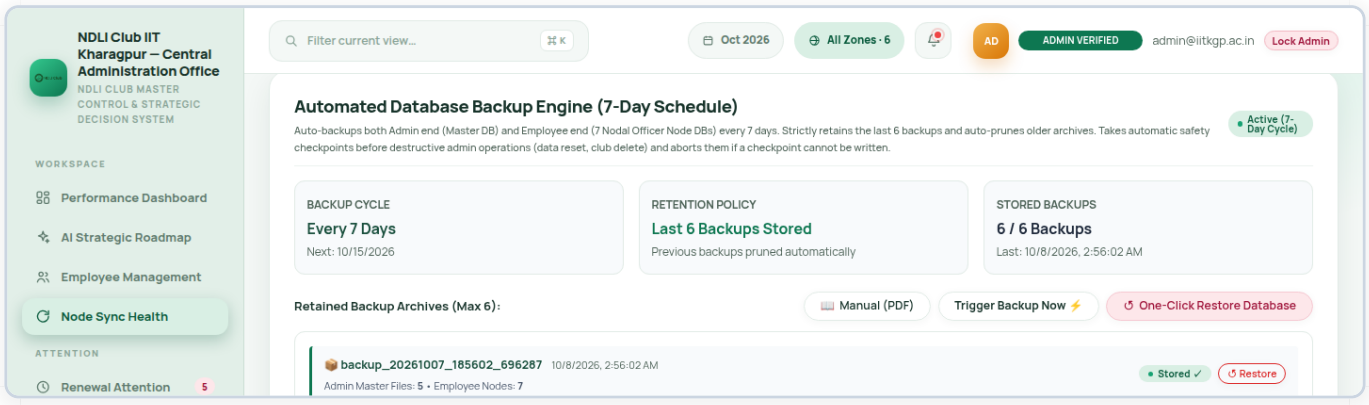


Figure 7.6: Google Drive & Node Sync Health Tab — 7-Day Rolling Backup Status & Cloud Webhook Relay

- **Autonomous 7-Day Cadence:** Archives both Central Master DBs and all 7 Regional Node DBs into compressed ZIP snapshots.
- **Strict 6-Version Retention Invariant:** Preserves only the last 6 historical snapshots (`MAX_RETAINED_BACKUPS = 6`). All older archives are automatically pruned, preventing storage bloat.
- **Google Drive Webhook Relay:** Every backup automatically mirrors a timestamped ZIP archive directly into the Google Drive `backups/` repository via Webhook Relay.
- **Signature Files Included:** the authorized signatory signature assets are bundled inside every Drive archive, so a restored backup is immediately certificate-ready.
- **Point-in-Time Drive Copies:** every local backup event — including the automatic checkpoint taken just before a destructive operation (test-data wipe, club deletion, restore) — immediately attempts its own Google Drive copy. The backup console reports the outcome in the `drive_sync` status field ("SUCCESS" only when the Drive relay confirmed receipt; "FAILED" otherwise).
- **Troubleshooting:** relay or sync failures are written to `sync_issues.log` in the data directory together with a loud entry in the server log — an empty log with "SUCCESS" statuses is the healthy state.

7.9 One-Click Emergency Database Restoration & Codebase Scripts ENTERPRISE

> ⚠️ **Critical Operational Procedure**
> The One-Click Emergency Restore operation **completely overwrites both Central Master DBs and all 7 Regional Node DBs** with data from the selected backup archive. Access is strictly gated by a **Dual-Layer Security Protocol**.

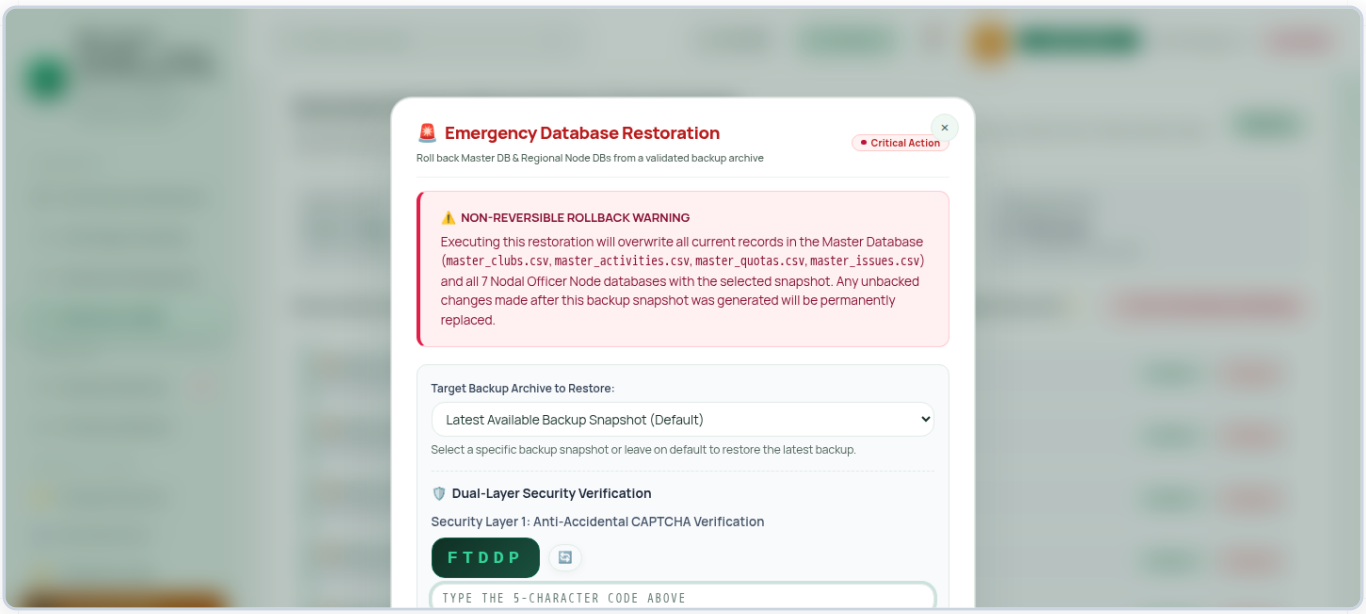


Figure 7.7: Emergency Database Restore Security Modal — Dual-Layer Challenge (Anti-Bot CAPTCHA + Master Admin Password)

The Dual-Layer Security Architecture

LAYER 1 **Dynamic Anti-Bot CAPTCHA Challenge**

Generates an unforgeable, randomized 5-character alphanumeric token rendered directly in an isolated HTML5 Canvas and validated server-side.

LAYER 2 **Master Administrator Password Re-Authentication**

Executing database rollback strictly requires re-authenticating as an active administrator: the entered credential is verified against the stored PBKDF2-HMAC-SHA256 hash of an ACTIVE admin account — no legacy, fallback, or pre-set credential of any kind is accepted — defeating session hijacking. Restore re-authentication attempts count toward the same brute-force lockout as sign-in (§5.1).

> 🔁 **Restore Semantics — What "Restore" Actually Does**
> The selected archive replaces **both** Central Master DBs and all Regional Node DBs wholesale — a full state replacement, not a merge. Before the overwrite, the system automatically takes a **pre-restore safety checkpoint** (undo point); if that checkpoint cannot be written, the restoration is aborted so current data is never left unprotected. Signature assets travel inside the archive (§7.8), so a restored system is immediately certificate-ready.

> 🛠️ **Codebase-Level One-Click Emergency Restore Utilities**
> For code-level emergency recovery, the system provides one-click restore scripts: `restore_to_stable_v2.bat` and `python restore_stable_v2.py`, instantly rolling back the application to the certified Most Stable Version 2 (`stable-v2` , commit `3f864f1`).

7.10 Employee Management, Certificates & Public QR Verification

7.10.1 Regional Employee Account Lifecycle Management

The Central Administration Portal provides centralized control over all Regional Officer accounts:

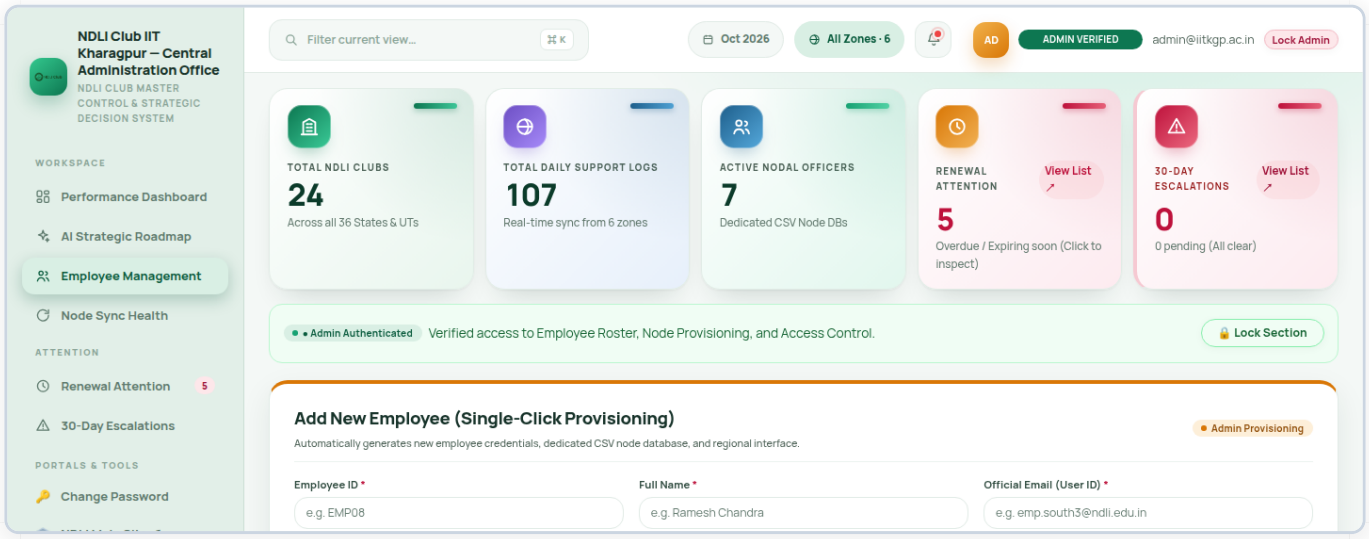


Figure 7.8: Central Administration — Regional Employee Roster, Access Toggles & Account Provisioning Form

- **Instant Provisioning:** Enter full name, official email, assigned zone, and covered states to generate a new officer node directory (e.g., `emp08`), dedicated CSV databases, and credentials.
- **Access Modification & Blocking:** Single-click **"🚫 Block"** button immediately suspends an officer's access without deleting their historical performance quota or support activity logs.
- **Direct Node Impersonation:** Click **"Open UI ↗"** to launch the employee's portal view in a new tab for quality audits and administrative verification.

7.10.2 Dynamic Renewal Certificate Customization

Administrators can configure the layout and visual assets of official NDLI Club Verification Certificates, including authorized signatory designations, high-resolution PNG signatures, and the official IIT Kharagpur institutional seal:

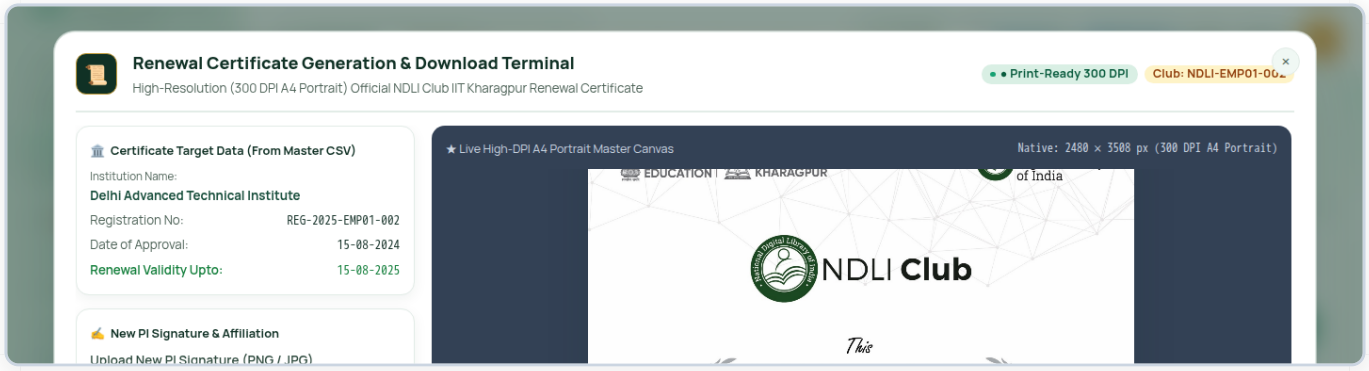


Figure 7.9: Dynamic Renewal Certificate Engine — Institutional Seals, Signatory Vectors & Sutradhar Typography

> 📌 **Employee Self-Service (client requirement, 2026-09-26)**
> Regional officers generate renewal certificates from their own portal and manage the Principal-Investigator signature themselves — the complete flow is documented in §6.8. Every signature upload is validated server-side (genuine PNG/JPEG image data only, checked by magic bytes, with a hard 2 MB cap) and every change is attributed to the signed-in officer. Admin-side layout and asset configuration above remains available for standardization. Every certificate — officer- or admin-generated — carries the public verification QR described in §7.10.3.

7.10.3 Public Certificate Verification (QR) V4.1 FEATURE

Every renewal certificate generated in the portal — by a regional officer (§6.8) or by the Central Office (§7.10.2) — embeds a **publicly verifiable QR code** in its print layout. Scanning it opens the public verification page: no login, no account. The page authenticates the certificate in real time and shows the club's current registration status, so any third party holding the printed certificate — an institution, a funding body, a verification desk — can confirm it in seconds. The window is documented here exactly as deployed: it has been reviewed and confirmed fit for purpose, and no changes to it are proposed.

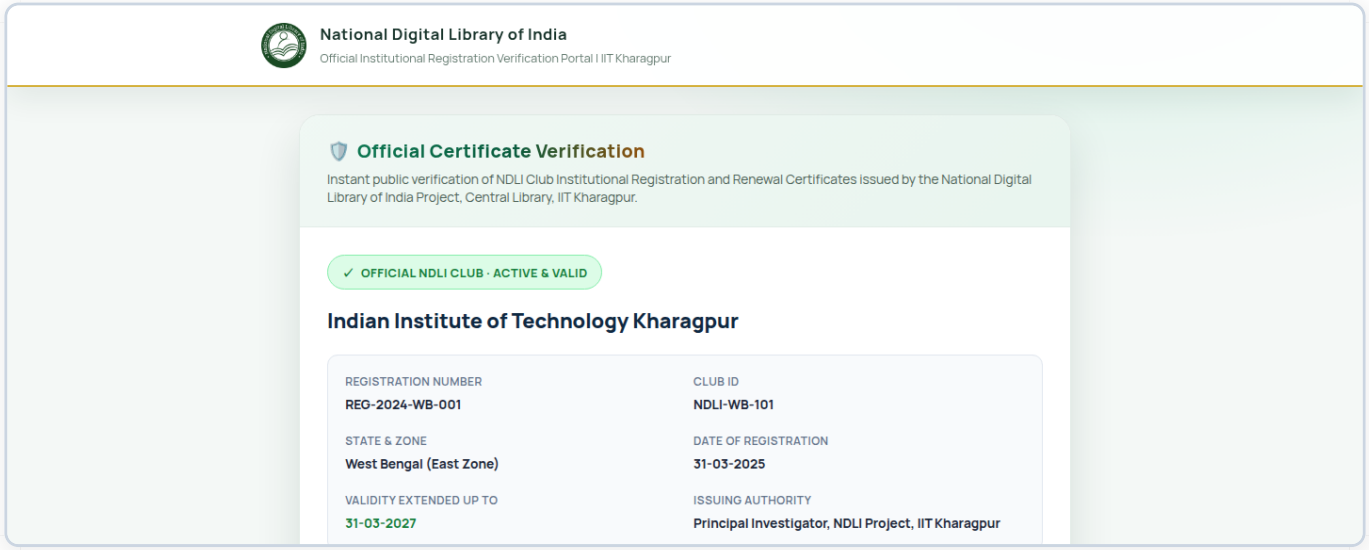


Figure 7.10: Public Certificate Verification Window — the Signed QR Result Card Presents Certificate Facts Only (Institution, Registration, Validity)

Why a QR Code on Paper Can Be Trusted

- **Minted and signed server-side:** the QR encodes a token of the form `v1.<key_id>.<payload>.<signature>`, signed with HMAC-SHA256 by a server-managed signing key. The operator controls that key (via the `NDLI_QR_SIGNING_KEY` environment variable); otherwise a strong key is generated once and held with restrictive file permissions — there is no repository-visible fallback, so a token is never forgeable from public sources.
- **Anti-forgery:** a valid signature proves the certificate came from this system for that exact registration number. Any alteration breaks the signature, and the page reports that the certificate *could not be authenticated as an official NDLI certificate*.
- **Anti-enumeration:** the public page is not a lookup service — it deliberately offers no search facility. It verifies only the certificate you hold in your hand; an unknown registration number returns *"No registered NDLI Club matches this certificate's registration number"*.
- **Built to last on paper:** tokens carry no expiry, because printed certificates must keep verifying for years. Key rotation is supported — each token names its signing key (`key_id`), so certificates issued earlier remain attributable after the operator rotates the signing key.
- **Who may mint:** tokens are minted through an authenticated portal API, and only for an exactly registered club. An administrator may mint for any club; a regional officer only for clubs inside their own node.

What the Verified Card Shows

The result card presents certificate facts only: institution name, registration number, club ID, state & zone, date of approval, validity status (**ACTIVE & VALID** or **EXPIRED — RENEWAL DUE**), and a verified-at timestamp. It never exposes patron/president/secretary contact details or any internal record (§9.3). If the verification service is temporarily unreachable, the page says so plainly instead of guessing.

7.11 Modern Minimalist UI & Complementary Navigation System ENTERPRISE

The user interface incorporates an ergonomic design system across all three portals (Master Landing, Employee Workspace, and Admin Control Center). The portal shell follows the approved Sample-C layout — a fixed left sidebar carrying the brand and sectioned navigation (with active states and count badges) plus a slim topbar above a roomy content column; every colour, spacing step, corner radius and elevation level comes from one shared **design-token** set, the **R21 dashboard pass** extends that same language into the analytics surfaces, feedback is delivered through **toast notifications**, and the floating **Robu assistant** launcher (§6.7) sits in the corner of both working portals:

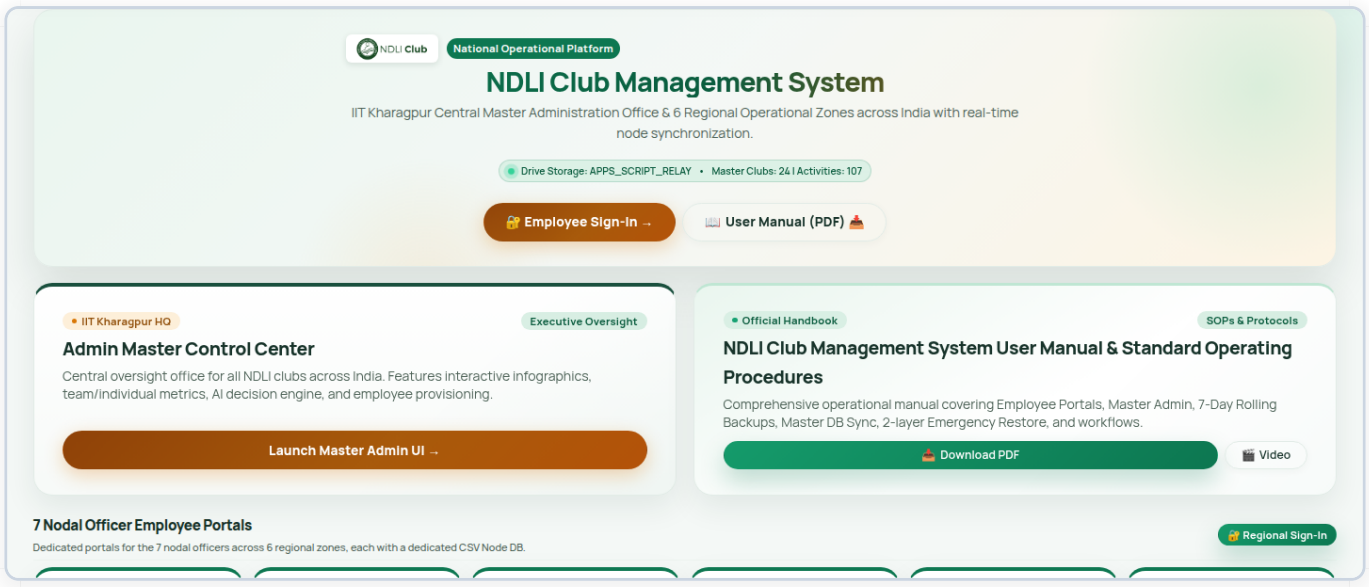


Figure 7.11: Modern Minimalist Landing Gateway — Manrope Typography, Amber Accent CTAs & Staggered Portal Cards

📊 R21 Dashboard Modernisation — Gradient Infographics & the Activity Distribution Matrix

Gradient KPI suite: tinted gradient KPI cards with floating accent pills, saturated gradient icon chips carrying white glyphs, urgency-tinted metric values and gradient-clipped card titles. **Signature chart set:** brand-gradient donut segments with legend mini-bars, a smooth Catmull-Rom monthly trajectory (gradient line, glow, dashed grid), renewal-ring gradients, a segmented pill gauge and gradient support tiles. **Activity Distribution Matrix:** a contribution share ribbon above heat-tinted per-officer rows — rank, officer name with zone chip, Clubs cell, Support cell, Share pill and a Last-Activity recency chip.

Design System Tokens & Architectural Elements

🔤 Manrope Typography

Google Fonts *Manrope* (geometric sans-serif with superior legibility) paired with *Inter* for dense numerical tables and data matrices.

🎨 Complementary Palette & Amber CTAs

High-contrast complementary color styling on navbar action buttons ("Employee Login", "Employee Portals") and warm amber CTAs for immediate visual recognition.

🪟 Glass Morphism Surfaces

Master lock card and modal overlays utilize deep-frosted glass effects with `backdrop-filter: blur(28px)`, ambient diffusion, and multi-layered borders.

📱 Zero-Scroll Login & Chip Equalization

Compact form heights and equal-width officer chip grids eliminate vertical scrolling on standard viewports, maximizing active workspace visibility.

🌊 Motion System: Spring Physics & Scroll-Reveals

Modals open with spring physics (`cubic-bezier(0.34, 1.56, 0.64, 1)`), buttons ripple on click, and cards cascade in via an `IntersectionObserver` hook — one consistent motion language across both portals.

🔔 Toasts, Focus States & Accessibility

Save/delete confirmations arrive as non-blocking toast notifications (auto-dismiss, action-aware colours). Every interactive element shows a visible focus ring (`ui-focusable`) for keyboard users, dialogs carry ARIA labels and focus traps, and status changes are announced to screen readers.

8. Maintenance, Enterprise 50k Scaling & FAQs

⚡ **Certified Enterprise Capacity Benchmarks (50,000+ Clubs & 200 Concurrency)**

Primary Key Lookups: 0.64ms avg (O(1) hash index across 50,000 records). **New Inserts:** 13.03ms fast-path O(1) append (zero full-file rewrite). **200 Concurrent Tasks:** Handled in 179.2ms (0 errors). **Crash Shield:** 10-tier Windows lock backoff & global UI error boundary.

8.1 File & Directory Topology

```
ndli_club_management/
├── app.py                    # Threaded HTTP Server & REST API (512 slots, 35+ endpoints)
├── auth.py                   # PBKDF2 Password Manager & Session Token Engine
├── config.py                 # Global Ports, Schemas & Scalability Configuration
├── restore_stable_v2.py      # One-Click Codebase Restore Utility (stable-v2)
├── restore_to_stable_v2.bat  # Windows One-Click Codebase Restore Batch
├── db/
│   ├── csv_engine.py         # O(1) Key Hash Indexing, Mutex Locking & 10-Tier Retry
│   ├── sync_engine.py        # Node-to-Master Synchronization & Fast ISO Parser
│   ├── storage_adapter.py    # Local & Bounded Drive Sync (_UPLOAD_SEMAPHORE=4)
│   └── backup_engine.py      # 7-Day Scheduler, ZIP Archiver & Retention Safeguard
├── tests/
│   ├── test_scaling_benchmark.py          # 50,000 Club Records Benchmark
│   └── test_scale_50k_and_concurrency.py  # 200 Concurrent Requests Stress Test
├── docs/
│   ├── user_manual.html      # This document (HTML source)
│   └── NDLI_Club_Management_User_Manual.pdf  # Compiled 24-Page User Manual
└── templates/ & deployment_gas/  # Dual-Deployment 100% Synced Portals
```

8.2 Operational Frequently Asked Questions (FAQs)

Q: How does the system scale to 50,000+ clubs and 200 real-time requests?

The CSV engine builds in-memory hash indices (`key_indices`) for O(1) sub-millisecond lookups, appends new rows directly in O(1) (~13ms) without rewriting multi-megabyte files, and handles 200 concurrent HTTP requests via bounded thread queues in <180ms.

Q: What is the function of the "Run Full Node-to-Master Reconciliation" button?

It audits all regional employee nodes, recovers missing records logged offline, normalizes schemas, and recomputes all quota tallies and star ratings to guarantee 100% data integrity without data loss.

Q: In an emergency, how do I restore the database or codebase?

For database: click **"One-Click Restore Database ↺"** in the Admin Portal (requires the anti-bot CAPTCHA plus re-authentication with an active administrator credential — §7.9). For codebase rollback: run `restore_to_stable_v2.bat` or `python restore_stable_v2.py` to return instantly to certified stable-v2.

Q: Sign-in says "Too many failed sign-in attempts" — what now?

8 failed attempts inside 10 minutes pause sign-in for 5 minutes (§5.1). Wait out the cool-off and sign in again. If the credential itself is lost or expired, contact the Central Administration Office — use only the credential issued to you by your administrator.

Q: How can a third party verify a printed renewal certificate?

Scan the QR code printed on it. The public verification page (§7.10.3) authenticates the certificate against its signed token and shows only certificate facts — institution, registration number, and validity. There is no search facility and no personal data on the page.

Q: Where is our data stored, and who can see it?

Role-scoped by design: officers see only their own node, the Central Office sees the national picture. Storage, retention, and assistant-logging details are set out in §9 Privacy & Data Handling.

9. Privacy & Data Handling

This charter states plainly what the platform stores, why it is collected, who may see it, and how long it is kept. It covers both portals, the Robu assistant, and the public verification window — and it is written to be read, not to be legal cover.

9.1 What the System Stores

- **Club registry:** institution identity plus stakeholder contacts (patron/president/secretary emails), approval dates, and renewal dates.
- **Employee accounts:** name, official email, role, zone, and assigned states. Credentials are never stored in readable form — PBKDF2-HMAC-SHA256 (100,000 rounds) with per-user random salts — and session tokens are random values that expire with the browser tab.
- **Operational records:** support activity logs (institution + notes + timestamp), issue/reminder records, and quota tallies.

9.2 Consent & Purpose

- Stakeholder contact details are collected solely for NDLI Club coordination: approvals, renewals, certificate dispatch, and support follow-up. They are not sold and not shared beyond NDLI / IIT Kharagpur operations.
- Support activity notes describe institutional engagement — keep them professional and factual, and avoid recording personal data of individual students.

9.3 Who Can See What

- **Role-scoped by design:** an employee sees ONLY their own node's clubs, activities, and issues; the Central Administration Office sees the aggregate national picture (§6.7).
- **Public verification window:** exposes ONLY certificate facts — institution, registration number, validity (§7.10.3) — never contacts or internal records.
- **Account administration:** provisioning and blocking of employee accounts is a Central Office action, and blocking preserves historical logs for audit.

9.4 Backups, Retention & Deletion

- **Rolling backups:** automated 7-day backups retain a maximum of 6 archives (older ones pruned) and are mirrored to the institution's Google Drive. Signature files are bundled so a restore is certificate-ready (§7.8).
- **Point-in-time copies:** Drive copies are taken automatically before destructive operations (test-data wipe, club deletion, restore). Every such event records its `drive_sync` outcome ("SUCCESS"/"FAILED"), and failures are written to `sync_issues.log` (§7.8).
- **History is preserved:** blocking an employee account never deletes historical activity logs — auditability comes first.

9.5 Assistant (Robu) Interactions

Questions and 👍/👎 feedback are logged to improve future answers. Robu is strictly read-only and role-scoped (§6.7); the interaction audit trail exists for quality and safety review only.

9.6 Certificates & Signatures

The Principal-Investigator signature image is uploaded and managed by the regional officer (§6.8); every upload is validated (genuine image magic bytes, 2 MB cap) and attributed. Public QR verification proves authenticity without exposing any personal data (§7.10.3).

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